

Ikram Aloui

Cloud-Native Security Engineer

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SUMMARY

Cloud-native security engineer with an M.Sc. in Information Systems Security, working on Harbor, the CNCF-graduated container registry. Writes production Go that upstream maintainers reviewed and merged: Cosign v3 bundle signature support released in Harbor v2.15.0, and two security fixes backported to the v2.15.0 release branch. Built the security audit logging system for Harbor Satellite, the project's edge companion, as a CNCF LFX Mentorship deliverable.

TECHNICAL SKILLS

Supply Chain Security: Cosign, Sigstore, SBOM (CycloneDX, SPDX), VEX, Syft, Trivy, attestations, vulnerability disclosure, CodeQL
Cloud Native: Kubernetes, K3s, RKE2, Docker, Helm, Rancher, containerd, OCI, Harbor, Zot
Identity & Zero Trust: SPIFFE/SPIRE, mTLS, Istio, Cilium/eBPF, cert-manager, OIDC, RBAC
Languages: Go, Python, TypeScript/JavaScript, Java, C, SQL/PLSQL
Cloud & CI/CD: AWS (EKS, RDS, S3, IRSA), OpenStack, GitHub Actions, GitLab, Ansible
Observability: Prometheus, Grafana, Jaeger, Kiali, OpenTelemetry, syslog/RFC 5424, Wazuh, Splunk
AI & Agent Tooling: Claude Code, Gemini, Codex; multi-agent workflows, agent task design and evaluation, AI code review

EXPERIENCE

Software Engineer (Part-time) · 8gears AG Oct 2025 – Present
Zurich, Switzerland (Remote) · Intern until Jul 2026

Security engineering across Harbor core, Harbor Satellite, and internal tooling: supply-chain security, vulnerability discovery and remediation, and hardening of the registry and its release infrastructure. Work ships as production Go, reviewed and merged upstream, with fixes backported to supported release branches.

LFX Mentee, Harbor Satellite · The Linux Foundation / CNCF Mar 2026 – May 2026
Remote · Competitive paid CNCF mentorship, Term 1, 2026

- Designed and implemented **security audit logging** end to end in Go across Ground Control and Satellite: canonical JSON event schema, a pluggable transport interface, and syslog (RFC 5424) and OpenTelemetry (OTLP/HTTP) transports.
- Verified the pipeline against live **Wazuh and Splunk** deployments.
- Authored the project's K3s reference architecture and SPIFFE/SPIRE deployment guides, and the Hugo blog infrastructure that publishes them.

Cybersecurity Intern · CERIST Feb 2025 – Jul 2025
Algiers, Algeria · M.Sc. thesis project

- Built a **Zero Trust architecture** for cloud-native workloads: Kubernetes micro-segmentation with Cilium/eBPF and mTLS through Istio.
- Automated the DevSecOps chain: Harbor, Trivy, and Cosign in CI/CD, certificate rotation via cert-manager, and multi-cluster administration through Rancher.
- Added real-time observability with Prometheus, Jaeger, and Kiali alongside dynamic network policies.

SELECTED OPEN SOURCE CONTRIBUTIONS

Cosign v3 bundle signature format, shipped in Harbor v2.15.0 goharbor/harbor #22628
Credential exfiltration fix in registry ping, backported to release-2.15.0 with a regression test goharbor/harbor #23463

- Cosign keyless signing for Harbor releases**, every release from v2.15.0 onward ships a verifiable signature bundle [goharbor/harbor #22578](#)
- Security audit logging with syslog and OpenTelemetry**, delivered as the CNCF LFX Mentorship project [harbor-satellite #448](#)
- Process crash fix in the audit event resolver**, backported to release-2.15.0 [goharbor/harbor #23461](#)

EDUCATION

- M.Sc., Information Systems Security · USTHB, Algiers** Sep 2023 – Jul 2025
Intrusion detection, cloud security, Zero Trust architecture, secure system administration.
- B.Sc., Computer Science · USTHB, Algiers** Sep 2020 – Jun 2023

WRITING & APPLIED RESEARCH

- Hardening Harbor on AWS: Zero-Static-Secret Architecture** ([container-registry.com](#)). RDS IAM Authentication and IRSA, removing every long-lived credential from a Harbor deployment.
- K3s + Harbor Satellite at the Edge** and **Deploying Harbor Satellite with SPIFFE/SPIRE** ([satellite.container-registry.com](#)). Reference architecture and step-by-step deployment guides.
- Implementing a Secure Software Supply Chain** (technical report, 2025). An RKE2, Longhorn, and Harbor platform with Trivy scanning and Cosign signing, proven when Harbor's policy blocked a vulnerable image from being pulled into the cluster.
- From Blind Spots to Full Visibility** (CERIST technical report, 2025). Zero-code Istio observability for an 11-service application; a simulated 2,000+ req/s DDoS was detected within 60 seconds by a custom Prometheus alerting rule.
- Hands-On Exploration of Bill of Behaviour** (technical report, 2025). Runtime behavioural profiling on Kubernetes, catching a live command injection that SBOM-only static analysis misses.

CERTIFICATIONS & LANGUAGES

- Microsoft Azure Fundamentals (2024) · Ethical Hacking: Cloud Computing, Cybrary (2024) · TryHackMe Advent of Cyber (2023)
- Arabic (native) · French (proficient) · English (fluent)